



MUKESH SETHY

ROBOTICS ENGINEER

PROFILE

Accomplished Mechatronics Engineer with seven and half years of specialized experience in robotics, electric vehicle technologies, and embedded systems. With a strong academic background in Mechatronics, Automation and Robotics, I have developed sophisticated software solutions, managed complex ECU development, and demonstrated expertise in thermal system integration for electric bus technologies. My professional journey includes comprehensive work with Daimler Trucks and Mercedes Benz Research and Development, where I have excelled in data analytics, control system design, and innovative technological implementations.

WORK EXPERIENCE

Rapyuta Robotics, Chennai

JUNE 2025 - PRESENT

Electrical Engineer

- Currently working as a core mechatronics development engineer for Rapyuta Robotics ASRS robot.
- Introduced Project Lifecycle Management tool and process for streamlining Bill of Materials handling for SCM and Engineering.
- Worked on multiple site issue identification and RCA for debugging.

Daimler Trucks Innovation Centre India - Split from Mercedes Benz Research and Development India - Bengaluru

Senior Engineer

AUGUST 2018 - JUNE 2025

- Component Responsible (ECU Owner) for Cooling Control Unit ECU.
- Managed new ECU lifecycle from specification development to Start of Production.
- Designed ECU I/O wiring harness for CAN, sensors and actuator communications.
- Supported Cybersecurity (TARA) and Functional Safety (HARA) assessments.
- Performed root cause analysis for production and component issues.
- Implemented country and customer specific cooling system requirements.
- Built a thermal test bench with actual bus components for validation.

Engineer - Data Analytics for e-Citaro(electric Bus)

- Imported and processed logged system data from e-Citaro dataloggers in field vehicles for system functionality monitoring.
- Developed a Python-based GUI for CAN signal data visualization.
- Monitored the functionality of cooling and HVAC systems for high-voltage components.
- Supported MIL/SIL/HIL testing for Cooling System application software development.

CONTACT

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Chennai

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EDUCATION

2013-2018

N.I.T ROURKELA

- Dual Degree M.Tech in Mechatronics and Automation and B.Tech in Mechanical Engineering

2010-2012

D.A.V PUBLIC SCHOOL,
UNIT-8, BHUBANESWAR

- 12th, CBSE

TECHNICAL SKILLS

- Python
- MATLAB-Simulink
- Vector CANoe
- Vector CANape
- Arduino
- Solidworks
- Eagle CAD
- ROS 2

LANGUAGES

- English (Fluent)
- Hindi (Fluent)
- Odia (Fluent)
- German (Basics)

ACADEMIC PROJECTS

- **Path Planning for NAO Humanoid Robot** 2017-2018
 - Master Thesis: Developed AI-based robot localization, obstacle mapping, and path planning using RGBD camera data.
- **Automatic Launch Control System** 2016
 - Designed a control system to reduce wheel slippage during vehicle launch using CAN Bus commands.
- **Pneumatic Gear Shifting System** 2015-2016
 - Designed a semi-automatic gear-shifting system controlled by microcontroller and ECU.
- **Formula SAE: Data Acquisition System** 2015-2016
 - Designed a semi-automatic gear-shifting system controlled by microcontroller and ECU.
- **Balloon Satellite Telemetry and Tracking** 2014
 - Led electronics team to build a satellite carrying sensors and cameras for atmospheric data logging and recovery.

PUBLICATIONS AND PATENT

- **Publications:**
 - "A Sensor-based Computer Vision Approach for Humanoid Navigation" – ICN:3I-2017, IIT Roorkee.
 - "Humanoid Navigation: An Intelligent Computer Vision-Based Approach" – IEEECON2018.
 - "An Intelligent Computer Vision Integrated Regression-Based Navigation Approach for Humanoids."
- **Patent:**
 - System for Optimal Management of Energy Consumption in Electric Vehicles (Application Number: 202211025514).

EXTRA CURRICULAR ACTIVITIES

- **Captain** 2014-2015
CYBORG - Robotics and Automation Club of NIT Rourkela
- **Electronics Head** 2014-2015
ECO-KART - Fabrication of a motor powered Go-Kart
- **Vice-Captain and Electronics Head** 2015-2017
FORMULA SAE - Fabrication of a Formula Student vehicle
- **First Position** 2014
IRC 2014, International Robotics Challenge (Zonal Round)